FlowMLLab | Week 1.2

# Pressure–Velocity Coupling

## SIMPLE, PISO and PIMPLE

Finite-volume equations and the lid-driven cavity

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**The central problem.** Momentum determines how fluid accelerates under pressure and viscous forces. Continuity constrains the resulting velocity. Pressure–velocity coupling obtains fields that satisfy both requirements on the same grid.

**Prerequisites.** Continuity and component momentum equations, finite differences, a linear system, and the Week 1 streamfunction–vorticity cavity example. Matrix notation is introduced after the scalar finite-volume equations.

**Learning outcomes.** Derive a cell pressure-correction equation; interpret predictor and corrected velocities; distinguish steady iteration, inner correction and outer correction; evaluate continuity, momentum defects, accuracy and computational work.


| Sections | Topics |
|---|---|
| 1–4 | Conservation laws, staggered grid, momentum coefficients |
| 5–8 | Pressure correction, conservation example and SIMPLE |
| 9–11 | Backward Euler, PISO and a numerical correction example |
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| 16–21 | Recorded cavity comparisons and coupling study |
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**Scope.** Two-dimensional, incompressible, laminar flow on a uniform staggered grid. The retained calculations compare $Re = 100, 400$ and $1000$. Higher-$Re$ spatial accuracy and transient accuracy require their own checks; convergence of a two-dimensional steady solver does not establish the physical stability of a three-dimensional flow.

# 1 Incompressible flow: conservation laws

Horizontal velocity $u_d(x_d, y_d, t_d)$, vertical velocity $v_d(x_d, y_d, t_d)$ and pressure $p_d(x_d, y_d, t_d)$ are the unknown fields. Subscript $d$ denotes dimensional quantities. For a Newtonian fluid with constant density $\rho$ and constant kinematic viscosity $\nu$, with no body force, continuity is

$$\frac{\partial u_d}{\partial x_d} + \frac{\partial v_d}{\partial y_d} = 0.$$

Conservation of horizontal and vertical momentum gives

$$\frac{\partial u_d}{\partial t_d} + u_d \frac{\partial u_d}{\partial x_d} + v_d \frac{\partial u_d}{\partial y_d} = -\frac{1}{\rho}\frac{\partial p_d}{\partial x_d} + \nu \left( \frac{\partial^2 u_d}{\partial x_d^2} + \frac{\partial^2 u_d}{\partial y_d^2} \right),$$
$$\frac{\partial v_d}{\partial t_d} + u_d \frac{\partial v_d}{\partial x_d} + v_d \frac{\partial v_d}{\partial y_d} = -\frac{1}{\rho}\frac{\partial p_d}{\partial y_d} + \nu \left( \frac{\partial^2 v_d}{\partial x_d^2} + \frac{\partial^2 v_d}{\partial y_d^2} \right).$$

The left sides contain local and convective acceleration. Pressure gradients and viscous diffusion provide the right-side accelerations. These two equations and continuity form the incompressible Navier–Stokes system.

**Scales.** Let $L$ be cavity side length and $U$ the lid speed. Define $x = x_d/L$, $y = y_d/L$, $t = t_d U/L$, $u = u_d/U$, $v = v_d/U$, and $p = (p_d - p_{\text{ref}})/(\rho U^2)$. Here $p_{\text{ref}}$ is a fixed arbitrary reference pressure. Then $Re = UL/\nu$ and

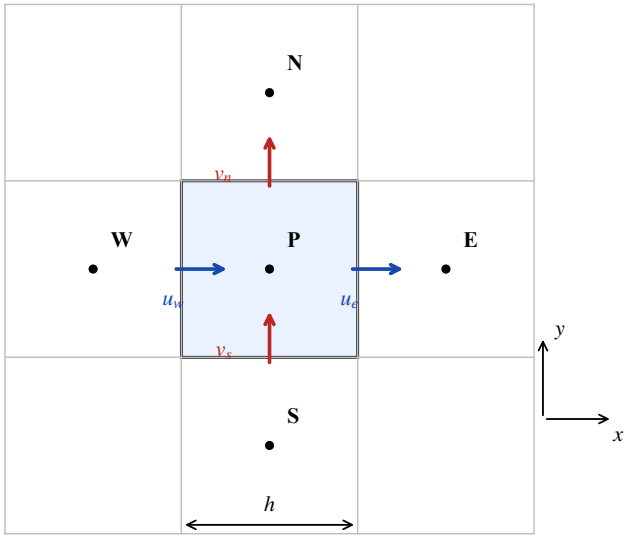
$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0,$$
$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \frac{1}{Re}\left( \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right),$$
$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{\partial p}{\partial y} + \frac{1}{Re}\left( \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right).$$

The following sections use these dimensionless quantities. At steady state the local time derivatives vanish, while convection, pressure and diffusion still balance.

**Cavity conditions.** The unit-square top lid has $u = 1, v = 0$; three stationary walls have $u = v = 0$. Every wall is impermeable. The ideal moving-lid corners contain a velocity discontinuity, so pressure and vorticity extrema there require care. The introductory case is $Re = 100$.

## 2 Staggered grid and cell mass balance

Pressure is stored at the center of each square pressure cell. Horizontal velocity is stored on vertical faces and vertical velocity on horizontal faces. This is the staggered MAC arrangement. The spacing is $h = 1/N$ for $N$ cells along one side.




Capital $P, E, W, N, S$ label pressure-cell centers; lowercase $e, w, n, s$ label the four faces of cell $P$. All horizontal velocity components are positive rightward; all vertical components are positive upward. The outward normal at the west or south face points in the negative coordinate direction.

Let $V$ denote the pressure-cell area $h^2$, $\partial V$ its perimeter, $\boldsymbol{u} = (u, v)$ the velocity vector, $\boldsymbol{n}$ its outward unit normal and $\mathrm{d}S$ a boundary-length element. Integrating continuity gives the net outward volume flux per unit depth:

$$\int_{\partial V} \boldsymbol{u} \cdot \boldsymbol{n} \, \mathrm{d}S = hu_e - hu_w + hv_n - hv_s = 0.$$

Define $r_{c,P}$ as the continuity residual of cell $P$: net outward volume flux divided by cell area. The subscript $c$ means continuity, and $P$ is a cell label. It is not a pressure change. Dividing by $h^2$ gives

$$\boxed{r_{c,P} = \frac{u_e - u_w + v_n - v_s}{h} = 0.}$$

Density is constant, so zero volume imbalance also means zero mass imbalance. A shared face enters one cell with a plus sign and its neighbor with a minus sign. Summing all cell balances cancels internal fluxes; an impermeable boundary has zero total outward flux.

**Why pressure is needed.** A guessed pressure can satisfy the momentum equations for a tentative velocity yet leave $r_{c,P} \neq 0$. The coupling algorithms adjust pressure gradients and face velocities together so that each cell satisfies continuity. Pressure is not prescribed independently throughout an incompressible cavity.

# 3 Finite-volume momentum at one velocity face

The velocity $u_e$ lies between pressure centers $P$ and $E$. Its momentum control volume is centered on that velocity face. Here $V_e$ is the velocity control volume, $\partial V_e$ its boundary, $\boldsymbol{n}$ the outward unit normal, $\mathrm{d}V$ an area element, and $\mathrm{d}S$ a boundary-length element, all per unit depth. Integrating horizontal momentum over this volume gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{V_e} u\,\mathrm{d}V + \int_{\partial V_e} u(\boldsymbol{u}\cdot\boldsymbol{n})\,\mathrm{d}S = -\int_{V_e}\frac{\partial p}{\partial x}\,\mathrm{d}V + \frac{1}{Re}\int_{\partial V_e}\nabla u\cdot\boldsymbol{n}\,\mathrm{d}S.$$

On the uniform interior grid, the volume-normalized pressure term is $-(p_E - p_P)/h$. Central interpolation approximates transported velocity at a momentum-volume face by the average of adjacent velocity samples. Central differences approximate its normal gradient.

**One interior row.** Denote the momentum control volume's neighboring horizontal velocities by $u_\mathrm{E}, u_\mathrm{W}, u_\mathrm{N}, u_\mathrm{S}$; these are different samples of the same horizontal component. Its equation is

$$\boxed{a_e u_e = a_\mathrm{E}u_\mathrm{E} + a_\mathrm{W}u_\mathrm{W} + a_\mathrm{N}u_\mathrm{N} + a_\mathrm{S}u_\mathrm{S} + b_e - \frac{p_E - p_P}{h}.}$$

Here $a_e$ is the center momentum coefficient, $a_\mathrm{NB}$ are neighbor coefficients, and $b_e$ contains known wall and time sources. All coefficients are normalized by control-volume area; no extra face area belongs in the pressure-gradient term.

Let $\beta = 1/(Re\,h^2)$. Let $c_\mathrm{E}, c_\mathrm{W}$ be the advecting horizontal velocities at the east/west boundaries of this momentum volume, and $c_\mathrm{N}, c_\mathrm{S}$ the advecting vertical velocities at its north/south boundaries. Central interpolation gives $c_\mathrm{E} = (u_e + u_\mathrm{E})/2$ and $c_\mathrm{W} = (u_\mathrm{W} + u_e)/2$. For the top and bottom boundaries,

$$c_\mathrm{N} = \frac{v_{n,P} + v_{n,E}}{2}, \qquad c_\mathrm{S} = \frac{v_{s,P} + v_{s,E}}{2}.$$

Here $v_{n,P}, v_{n,E}$ are the vertical velocities at the top faces of pressure cells $P, E$; $v_{s,P}, v_{s,E}$ are their bottom-face velocities. The interior central scheme uses

$$a_\mathrm{E} = \beta - \frac{c_\mathrm{E}}{2h}, \qquad a_\mathrm{W} = \beta + \frac{c_\mathrm{W}}{2h},$$
$$a_\mathrm{N} = \beta - \frac{c_\mathrm{N}}{2h}, \qquad a_\mathrm{S} = \beta + \frac{c_\mathrm{S}}{2h},$$
$$a_e = a_\mathrm{E} + a_\mathrm{W} + a_\mathrm{N} + a_\mathrm{S} + \frac{c_\mathrm{E} - c_\mathrm{W} + c_\mathrm{N} - c_\mathrm{S}}{h}.$$

For example, east-face convection contributes $c_\mathrm{E}(u_e + u_\mathrm{E})/(2h)$ on the left, and diffusion contributes $\beta(u_\mathrm{E} - u_e)$ on the right. Moving the convective neighbor term to the right leaves $(\beta - c_\mathrm{E}/(2h))u_\mathrm{E}$, which explains $a_\mathrm{E}$. Other faces follow from their outward-normal signs. Time and wall terms modify the diagonal and source. The vertical component has an analogous row.

# 4 Momentum coefficients and matrix notation

For a velocity location labeled $f$, let $\mathrm{NB}(f)$ be the neighboring velocity locations in its momentum row. The index $g$ labels one such location; $a_{fg}$ is the coefficient multiplying its velocity $q_g$. These are velocity-control-volume neighbors, rather than the pressure neighbors labeled $E, W, N, S$. Collect the source and neighbor contributions into

$$H_f = b_f + \sum_{g \in \mathrm{NB}(f)} a_{fg} q_g, \qquad d_f = \frac{1}{a_f}.$$

The symbol $q_g$ means a velocity degree of freedom; it is not volume flux. The scalar momentum row becomes

$$q_f = d_f \left[ H_f - (Gp)_f \right], \qquad (Gp)_e = \frac{p_E - p_P}{h}.$$

The vector $q$ collects all unknown interior horizontal and vertical face velocities. The scalar rows form two component linear systems, written together as

$$Aq = b - Gp.$$

The product $Dq$ collects the cell residuals: $(Dq)_P = r_{c,P}$. The product $Gp$ collects face pressure gradients. Multiplication by $d$ scales each face entry by its own $1/a_f$.

For $N$ cells per side, $p$ has $N^2$ entries and $q$ has $2N(N-1)$ unknown interior face velocities. Thus $A$ is $2N(N-1)$ square, $D$ has $N^2$ rows and $2N(N-1)$ columns, $G$ has the reverse shape, and $L = -DdG$ is $N^2$ square. The operator $\mathrm{diag}(a_f)$ places the coefficients on the diagonal and zeros elsewhere.

The diagonal of $A$ contains the center coefficients $a_f$; its off-diagonal entries are $-a_{fg}$. With $R = A - \mathrm{diag}(a_f)$ and $d = \mathrm{diag}(1/a_f)$,

$$H(q) = b - Rq, \qquad q = d\left[H(q) - Gp\right].$$

Thus $H$ is computed from current neighbor velocities; it is not another fluid property. Here $d$ is an inverse diagonal matrix, while $h$ is mesh spacing. In operator formulas $L$ denotes the pressure matrix; the $L$ used earlier in dimensional scaling denotes cavity length.

| Symbol | Meaning | Location or role |
|---|---|---|
| $p$ | Pressure relative to a reference | Cell centers |
| $q$ | Collected velocity components | Interior faces |
| $A, b$ | Momentum matrix and source | Two component systems |
| $d$ | Inverse center coefficients | Face-wise diagonal |
| $R$ | Off-diagonal part of $A$ | Neighbor coupling |
| $H = b - Rq$ | Source plus neighbor terms | Recomputed from $q$ |
| $D$ | Discrete divergence | Faces to cell balance |
| $G$ | Discrete pressure gradient | Centers to faces |
| $L = -DdG$ | Pressure operator | Cell-center system |

The notation abbreviates the scalar equations; it does not introduce additional physics. Each occurrence of $A, b, d$ refers to the current assembly, including any wall, transient or relaxation modifications.

# 5 Pressure correction from the momentum equations

A star denotes a tentative velocity. The superscript $k$ labels the current steady coupling iteration, not a power or a physical time level. Given the current pressure estimate $p^k$, solve momentum for $q^*$ while holding its coefficients fixed:

$$Aq^* = b - Gp^k.$$

**Pressure correction and pressure differences.** Let $p_{\text{corr}}$ be the pressure targeted by the current correction solve. Define

$$p' = p_{\text{corr}} - p^k, \qquad q' = q_{\text{corr}} - q^*.$$

A prime is an additive correction, not a derivative. The unrelaxed pressure change at that cell is

$$\boxed{\Delta p = p_{\text{corr}} - p^k = p'.}$$

It is a change at a fixed cell during coupling. By contrast, $p_E - p_P$ is a spatial pressure difference between neighboring cells at the same iteration. Neither is the continuity residual $r_{c,P}$.

Write $q_{\text{corr}} = q^* + q'$ and $p_{\text{corr}} = p^k + p'$. Substituting these into the same momentum system and subtracting the predictor gives

$$Aq' = -Gp'.$$

At one horizontal face this is

$$a_e u'_e = \sum_{g \in \text{NB}(e)} a_{eg} q'_g - \frac{p'_E - p'_P}{h}.$$

The exact correction couples neighboring velocity corrections. SIMPLE obtains a local correction by neglecting that neighbor-correction sum during this correction solve:

$$\boxed{u'_e = -d_e \frac{p'_E - p'_P}{h}}, \qquad d_e = 1/a_e.$$

Likewise,

$$u'_w = -d_w \frac{p'_P - p'_W}{h},$$
$$v'_n = -d_n \frac{p'_N - p'_P}{h}, \qquad v'_s = -d_s \frac{p'_P - p'_S}{h}.$$

The corrected velocity must satisfy cell continuity. Substituting the four face corrections gives

$$0 = r^*_{c,P} + \frac{u'_e - u'_w + v'_n - v'_s}{h}, \qquad r^*_{c,P} = \frac{u^*_e - u^*_w + v^*_n - v^*_s}{h}.$$

This equation determines pressure corrections from the predictor's cell imbalance. Neglecting neighbor corrections is an approximation to momentum coupling, not permission to violate continuity. Repeated coupling iterations are required to recover the target steady momentum solution.

# 6 The cell pressure equation and its boundary data

At each face, the corrected velocity is tentative velocity plus its correction:

$$u_e = u_e^* - \frac{d_e}{h}(p'_E - p'_P), \qquad u_w = u_w^* - \frac{d_w}{h}(p'_P - p'_W),$$
$$v_n = v_n^* - \frac{d_n}{h}(p'_N - p'_P), \qquad v_s = v_s^* - \frac{d_s}{h}(p'_P - p'_S).$$

Insert these into $(u_e - u_w + v_n - v_s)/h = 0$. The west and south terms change sign because their velocities enter cell outflow with minus signs:

$$0 = r^*_{c,P} - \frac{d_e}{h^2}(p'_E - p'_P) + \frac{d_w}{h^2}(p'_P - p'_W) - \frac{d_n}{h^2}(p'_N - p'_P) + \frac{d_s}{h^2}(p'_P - p'_S).$$

Collect the coefficients of each pressure correction and move the tentative residual to the right:

$$\frac{d_e + d_w + d_n + d_s}{h^2} p'_P - \frac{d_e p'_E + d_w p'_W + d_n p'_N + d_s p'_S}{h^2} = -r^*_{c,P}.$$

The center coefficient is the sum of four face weights; each neighbor has a negative coefficient. The right side is the negative cell divergence of the tentative velocity. The same face weight must be shared by its two adjacent pressure cells.

**Matrix form.** Applying $D$ to $q = q^* - dGp'$ gives

$$0 = Dq^* - DdGp',$$
$$L = -DdG,$$
$$Lp' = -Dq^*.$$

The positive-diagonal convention for $L$ explains the minus sign on the right. Once pressure is solved, the face correction $-dGp'$ cancels the computed cell imbalance, up to linear-solve error.

## 6 continued Boundary cells and pressure reference

**Impermeable walls.** The normal wall velocity is prescribed and is not an unknown interior face velocity. Its normal correction is zero. A prescribed wall-normal face contributes zero correction weight, so that face coefficient is omitted from the boundary-cell pressure row. Pressure assembly therefore has no correction flux across the external wall. For a cell adjacent to a stationary west wall, $u^*_w = u'_w = 0$, so its row is

$$\frac{d_e + d_n + d_s}{h^2} p'_P - \frac{d_e p'_E + d_n p'_N + d_s p'_S}{h^2} = -r^*_{c,P}.$$

There is no west pressure-neighbor term. At a corner, omit both prescribed wall-normal face weights. Zero normal correction does not mean zero pressure at every wall.

**Pressure reference.** The vector $\mathbf{1}$ contains one in every pressure entry. Adding a constant to every cell pressure changes no pressure gradient: $G\mathbf{1} = 0$. Consequently $L\mathbf{1} = 0$ and the closed-cavity pressure system has one arbitrary constant. Pinning means temporarily prescribing one pressure unknown, for example $p_{\text{ref cell}} = 0$, to make the linear solve unique. The implementation then removes the arithmetic cell mean:

$$\bar{p} = \frac{1}{N^2} \sum_P p_P, \qquad \widetilde{p}_P = p_P - \bar{p}.$$

The same constant is subtracted from every cell, so

$$\widetilde{p}_E - \widetilde{p}_P = (p_E - \bar{p}) - (p_P - \bar{p}) = p_E - p_P.$$

The gradient and corrected velocity are unchanged. The reported zero-mean pressure is a gauge choice, not an extra boundary force.

Tangential no-slip is enforced in momentum through wall diffusion. On the staggered grid it is distinct from the zero normal correction used in the cell mass balance.

# 7 Two-cell pressure-correction example

Consider two closed square cells with one shared vertical face. This is an algebraic conservation example, not a resolved cavity. The pressure guess is initially zero in both cells. Let $h = 0.5$, tentative shared velocity $u^* = 0.2$, and inverse momentum diagonal $d_e = 0.1$. Positive shared velocity leaves cell 1 and enters cell 2. All external normal velocities are zero.

$$r^*_{c,1} = \frac{0.2}{0.5} = 0.4, \qquad r^*_{c,2} = \frac{-0.2}{0.5} = -0.4.$$

The shared pressure weight is $d_e/h^2 = 0.1/0.25 = 0.4$. Therefore

$$0.4 \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} p'_1 \\ p'_2 \end{bmatrix} = \begin{bmatrix} -0.4 \\ 0.4 \end{bmatrix}.$$

The first row gives $p'_1 - p'_2 = -1$. With zero mean, $p'_1 + p'_2 = 0$, so $p'_1 = -0.5$ and $p'_2 = 0.5$. The corrected face velocity is

$$u = u^* - d_e \frac{p'_2 - p'_1}{h} = 0.2 - 0.1 \frac{0.5 - (-0.5)}{0.5} = 0.$$

Thus the unrelaxed changes are $\Delta p_1 = p'_1 = -0.5$ and $\Delta p_2 = p'_2 = 0.5$. Both corrected divergences are zero. A common constant added to $p'_1$ and $p'_2$ leaves this calculation unchanged.

**Pressure relaxation and velocity correction.** The pressure relaxation factor $\alpha_p$ is the fraction of $p'$ added to the pressure stored for the next momentum predictor: $p^{k+1} = p^k + \alpha_p p'$. The stored pressure change is therefore $\alpha_p p'$, rather than the full unrelaxed $\Delta p = p'$. If only $\alpha_p = 0.3$ of the velocity correction were applied, the shared velocity would remain $0.2 - 0.3(0.2) = 0.14$. The two mass defects would then be $+0.28$ and $-0.28$. With $\alpha_p = 0.3$, the stored pressures become $-0.15$ and $0.15$ from the zero initial guess, while the fully corrected shared velocity remains zero. This velocity need not yet satisfy momentum with the relaxed stored pressure; the next predictor and correction continue the coupling iteration. The conservative velocity correction is applied fully.

**Local and global conservation.** The initial global sum was already $0.4 - 0.4 = 0$, but each cell was imbalanced. A zero total boundary flux alone cannot certify local continuity.

# 8 SIMPLE: steady pressure–velocity coupling

SIMPLE means Semi-Implicit Method for Pressure-Linked Equations. Its index $k$ counts steady coupling iterations, not physical time. Each iteration assembles convection from the latest velocity, solves a momentum predictor, then solves the pressure-correction equation.

**Momentum under-relaxation.** Let $a_f^0$ be the original center coefficient of the current unrelaxed assembly. The superscript $0$ here identifies the coefficient before relaxation, not the initial physical time. The label rel means relaxed. The factor $\alpha_u$ controls momentum under-relaxation; for $0 < \alpha_u \leq 1$, the relaxed row uses

$$a_f^{\text{rel}} = \frac{a_f^0}{\alpha_u}, \qquad b_f^{\text{rel}} = b_f + \left(\frac{1}{\alpha_u} - 1\right) a_f^0 q_f^k.$$

The added source uses the original diagonal $a_f^0$. At a converged fixed point, $q_f^{k+1} = q_f^k$, the added diagonal and source contributions cancel. The target steady equation is unchanged. The pressure correction uses $d_f = 1/a_f^{\text{rel}}$.

**One steady iteration.**

$$\begin{aligned} A^{\text{rel}} q^* &= b^{\text{rel}} - G p^k, \\ L p' &= -D q^*, \\ q^{k+1} &= q^* - d G p', \\ p^{k+1} &= p^k + \alpha_p p', \qquad 0 < \alpha_p \leq 1. \end{aligned}$$

Velocity is corrected fully; only the stored pressure update is damped by $\alpha_p$. After boundary conditions and the pressure gauge are applied, the next iteration rebuilds the momentum coefficients from the corrected velocity.

**Stopping criteria.** Momentum relaxation can make updates small before the original equations are satisfied. The implementation checks the unrelaxed nonlinear steady momentum defect and the cell divergence. Both must meet tolerance. In the retained baseline, $\alpha_u = 0.7$ and $\alpha_p = 0.3$.

**Role.** SIMPLE is a natural steady solver. It can also be used inside a transient nonlinear solve, but the current implementation's SIMPLE branch has no physical time step. Its iteration count cannot be compared directly with the physical-step counts of PISO or PIMPLE.

# 9 Physical time and backward-Euler momentum

PISO and PIMPLE solve a transient momentum equation. Superscripts $n$ and $n+1$ denote physical time levels, separated by $\Delta t$. For one face velocity,

$$\left.\frac{\partial q_f}{\partial t}\right|_{n+1} \approx \frac{q_f^{n+1} - q_f^n}{\Delta t}.$$

If the spatial momentum row has center coefficient $a_f^{\text{space}}$ and source $b_f^{\text{space}}$, backward Euler produces

$$\left(a_f^{\text{space}} + \frac{1}{\Delta t}\right) q_f^{n+1} = \sum_g a_{fg} q_g^{n+1} + b_f^{\text{space}} + \frac{q_f^n}{\Delta t} - (Gp^{n+1})_f.$$

Thus the transient assembly adds $1/\Delta t$ to the diagonal and the known old velocity $q_f^n/\Delta t$ to the source.

**Old-time velocity is fixed.** At the beginning of the physical step, store $q^n$. Every momentum assembly and every correction within that step uses the same old-time source. Replacing $q^n$ with an intermediate corrected velocity would change the discrete physical problem.

**Nonlinear coefficients.** Convective coefficients depend on advecting velocity. They must be evaluated from a known estimate to form a linear predictor system. Such coefficients are called frozen or lagged during that solve. A perfectly solved linear predictor can still differ from the fully implicit nonlinear step if its coefficients were evaluated at an outdated velocity.

**Three iteration levels.** A linear solver solves one matrix equation. A pressure corrector couples pressure and velocity for a frozen momentum matrix. An outer corrector rebuilds that matrix using the latest velocity. None of these is an additional physical time advance.

Backward Euler is first-order accurate in time. Extra correctors can reduce algebraic coupling error, but do not change its time-discretization order. A larger stable $\Delta t$ still requires a temporal-accuracy assessment.

# 10 PISO: pressure corrections for a frozen momentum matrix

PISO means Pressure-Implicit with Splitting of Operators. At one physical step, assemble transient momentum once and solve

$$Aq^{*} = b - Gp^{\text{guess}}.$$

This predictor uses the pressure available from the preceding step or initial state. Its momentum residual for that pressure can be small, while its cell divergence is nonzero.

Set $q^{(0)} = q^{*}$. The index $m$ counts inner corrections within this one physical step. The component matrices $A$, their source $b$, inverse diagonal $d$, off-diagonal part $R$, and pressure operator $L = -DdG$ remain fixed throughout the inner loop. Neighbor velocities do change, so each correction computes

$$\begin{aligned} H^{(m)} &= b - Rq^{(m)}, \\ Lp^{(m+1)} &= -D\left(dH^{(m)}\right), \\ q^{(m+1)} &= d\left[H^{(m)} - Gp^{(m+1)}\right]. \end{aligned}$$

The pressure equation follows by applying $D$ to the last line and imposing $Dq^{(m+1)} = 0$. Unlike SIMPLE's $p'$, this implementation's $p^{(m+1)}$ is the full pressure field in the chosen gauge, rather than a pressure increment. It replaces the previous pressure; it is not added to it.

**Purpose of the next correction.** Reconstructed velocity changes the neighbor terms in $H$. Recomputing $H$ and solving pressure again improves the consistency of reconstructed velocity with the same frozen momentum equation. Repeating a linear pressure solve with an unchanged right-hand side is not an additional PISO correction.

Let $N_i$ be the specified number of inner corrections. After $N_i$ corrections, accept $q^{n+1} = q^{(N_i)}$ and $p^{n+1} = p^{(N_i)}$, then advance time once. The retained baseline has $N_i = 2$. No momentum or pressure relaxation is applied by this transient branch.

The nonlinear convection coefficients remain those used for the initial predictor. That limitation motivates an outer coefficient-update loop.

# 11 PISO: a frozen-matrix numerical example

This reduced two-velocity system illustrates coupling algebra; it is not a cavity grid. Let

$$A = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}, \quad b = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & -1 \end{bmatrix}, \quad G = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Here $p$ is one independent scalar pressure variable after removing an arbitrary common pressure offset. It is not a separate pressure value in each physical cell. The matrices are prescribed to illustrate coupling algebra; they are not derived from the earlier two-cell geometry. Continuity is $Dq = q_1 - q_2 = 0$. From $A$,

$$d = \begin{bmatrix} 1/4 & 0 \\ 0 & 1/3 \end{bmatrix}, \quad R = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad L = -DdG = 7/12.$$

Starting from $p^{\text{guess}} = 0$, the momentum predictor solves $Aq^* = b$:

$$q^* = \begin{bmatrix} 6/11 \\ -2/11 \end{bmatrix}, \qquad Dq^* = 8/11.$$

It satisfies momentum exactly but violates continuity. The first inner correction uses

$$H^{(0)} = b - Rq^* = \begin{bmatrix} 24/11 \\ -6/11 \end{bmatrix}, \qquad dH^{(0)} = \begin{bmatrix} 6/11 \\ -2/11 \end{bmatrix}.$$

Solve pressure and reconstruct velocity:

$$\frac{7}{12}p^{(1)} = -D(dH^{(0)}) = -\frac{8}{11}, \qquad p^{(1)} = -\frac{96}{77},$$
$$q_1^{(1)} = \frac{1}{4}\left(\frac{24}{11} + p^{(1)}\right) = \frac{18}{77}, \qquad q_2^{(1)} = \frac{1}{3}\left(-\frac{6}{11} - p^{(1)}\right) = \frac{18}{77}.$$

The signs in these two velocity formulas follow from $G = (-1, 1)^T$. Now $Dq^{(1)} = 0$. Nevertheless, substituting these fields into the original frozen momentum equation gives

$$r^{(1)} = Aq^{(1)} - b + Gp^{(1)} = \begin{bmatrix} 32/77 \\ -24/77 \end{bmatrix}.$$

The vector $r = Aq - b + Gp$ is the error in satisfying the momentum rows; exact satisfaction gives $r = 0$. Its infinity norm is the largest absolute entry, $\|r\|_\infty = \max_i |r_i|$, where $i$ labels a momentum row. Mass balance is exact here, but momentum consistency still needs improvement.

## 11 continued The second pressure correction

The corrected neighbor velocities change $H$, even though $A$ and $L$ are unchanged:

$$H^{(1)} = b - Rq^{(1)} = \begin{bmatrix} 136/77 \\ -18/77 \end{bmatrix}, \qquad dH^{(1)} = \begin{bmatrix} 34/77 \\ -6/77 \end{bmatrix}.$$

The next pressure and velocity are

$$\frac{7}{12}p^{(2)} = -\frac{40}{77}, \qquad p^{(2)} = -\frac{480}{539},$$
$$q^{(2)} = d\left[H^{(1)} - Gp^{(2)}\right] = \begin{bmatrix} 118/539 \\ 118/539 \end{bmatrix}.$$

The pressure is replaced by $p^{(2)}$; adding $p^{(2)}$ to $p^{(1)}$ would be incorrect. Again continuity is exact. The remaining frozen-momentum defect is

$$r^{(2)} = \begin{bmatrix} -8/539 \\ -8/539 \end{bmatrix}.$$

| Stage | $\lvert Dq\rvert$ | $\lVert Aq - b + Gp\rVert_\infty$ |
|---|---|---|
| Momentum predictor | $8/11$ | $0$ |
| Inner correction 1 | $0$ | $32/77 \approx 0.41558$ |
| Inner correction 2 | $0$ | $8/539 \approx 0.01484$ |

The exact constrained solution is $q_1 = q_2 = 2/9$ and $p = -8/9$. It follows by imposing $q_1 = q_2 = s$ in both original momentum rows: $5s - p = 2$ and $4s + p = 0$, hence $9s = 2$.

The second correction reduces momentum inconsistency in this example while preserving mass balance. This is a demonstrated property of these specified matrices, not a guarantee of monotone convergence for every CFD system.

The right-hand side of the pressure equation changes from $-8/11$ to $-40/77$. The example therefore distinguishes another PISO correction from another iteration of an unchanged linear pressure solve.

# 12 PIMPLE: outer coefficient updates and inner corrections

PIMPLE combines an outer momentum-update loop with PISO inner corrections. Physical time levels are $n, n+1$; $\ell$ labels outer updates and $m$ labels inner pressure corrections. The previous physical velocity $q^n$ is stored once.

At the start of outer pass $\ell$, assemble convection using the latest available corrected velocity $\widehat{q}^{(\ell)}$. With $I$ denoting the identity matrix, this gives new matrices and sources

$$A^{(\ell)} = A_{\text{space}}(\widehat{q}^{(\ell)}) + I/\Delta t, \qquad b^{(\ell)} = b^{(\ell)}_{\text{wall}} + q^n/\Delta t.$$

The same $q^n$ appears in every outer pass. Wall contributions may depend on the current assembly, but the prescribed wall motion and old-time source belong to the same physical step.

The first outer pass starts from $q^n, p^n$. The symbol $\widehat{p}^{(\ell)}$ denotes the latest available pressure at the start of outer pass $\ell$. A new momentum predictor is solved using that pressure. Then inner corrections hold the current outer assembly fixed:

$$\begin{aligned} A^{(\ell)} q^{(\ell,0)} &= b^{(\ell)} - G\widehat{p}^{(\ell)}, \\ H^{(\ell,m)} &= b^{(\ell)} - R^{(\ell)} q^{(\ell,m)}, \\ L^{(\ell)} p^{(\ell,m+1)} &= -D\left(d^{(\ell)} H^{(\ell,m)}\right), \\ q^{(\ell,m+1)} &= d^{(\ell)}\left[H^{(\ell,m)} - G p^{(\ell,m+1)}\right]. \end{aligned}$$

With $N_i$ inner corrections per outer pass, the initial and transfer rules are

$$\widehat{q}^{(1)} = q^n, \qquad \widehat{p}^{(1)} = p^n,$$
$$\widehat{q}^{(\ell+1)} = q^{(\ell,N_i)}, \qquad \widehat{p}^{(\ell+1)} = p^{(\ell,N_i)}.$$

Thus the final corrected fields of one outer pass initialize the next. Rebuilding $A$ also rebuilds $d$, $R$ and $L$. Each inner solve computes the full pressure field in the chosen gauge, as in this PISO implementation.

**Two different purposes.** Inner corrections improve pressure–velocity consistency for the current matrix. Outer passes update the nonlinear coefficients that matrix represents. Neither loop advances physical time. Only after the final outer pass are the accepted fields assigned to time $n+1$.

With one outer pass, this code reduces to its PISO algorithm. Its PIMPLE branch does not apply SIMPLE's under-relaxation. Additional loops reduce certain algebraic errors at extra cost; they do not increase backward-Euler's formal time order.

# 13 PIMPLE: a nonlinear coefficient example

Extend the preceding algebraic example by making its momentum coefficients depend on velocity:

$$A(s) = \begin{bmatrix} 4+s & 1 \\ 1 & 3+s \end{bmatrix}, \qquad s = (q_1 + q_2)/2, \qquad b = \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

Keep the same $D, G$ and initial $q = 0, p = 0$. At each outer pass, evaluate $s$ from its starting velocity, freeze $A(s)$, solve a new predictor, and perform two inner corrections. The source $b$ remains fixed. This toy coefficient law illustrates lagging; it is not a Navier–Stokes convection discretization.

At outer pass 1, $s = 0$, so the two inner corrections are exactly the previous PISO example. They end at $q_1 = q_2 = 0.2189239332$. The next outer pass uses

$$A^{(2)} = \begin{bmatrix} 4.2189239332 & 1 \\ 1 & 3.2189239332 \end{bmatrix},$$

and recomputes $d, R, L$ before the new predictor and inner corrections.

| Outer pass | Frozen $s$ | Final $q_1 = q_2$ | Final $p$ | Nonlinear defect |
|---|---|---|---|---|
| 1 | 0 | 0.2189239332 | –0.8905380334 | $3.30854 \times 10^{-2}$ |
| 2 | 0.2189239332 | 0.2119026446 | –0.8940486777 | $1.53537 \times 10^{-3}$ |
| 3 | 0.2119026446 | 0.2122289655 | –0.8938855173 | $7.14784 \times 10^{-5}$ |

The reported defect is the largest absolute component of the nonlinear momentum residual, evaluated with coefficients rebuilt from the final velocity:

$$\| A\left((q_1 + q_2)/2\right) q - b + Gp \|_\infty .$$

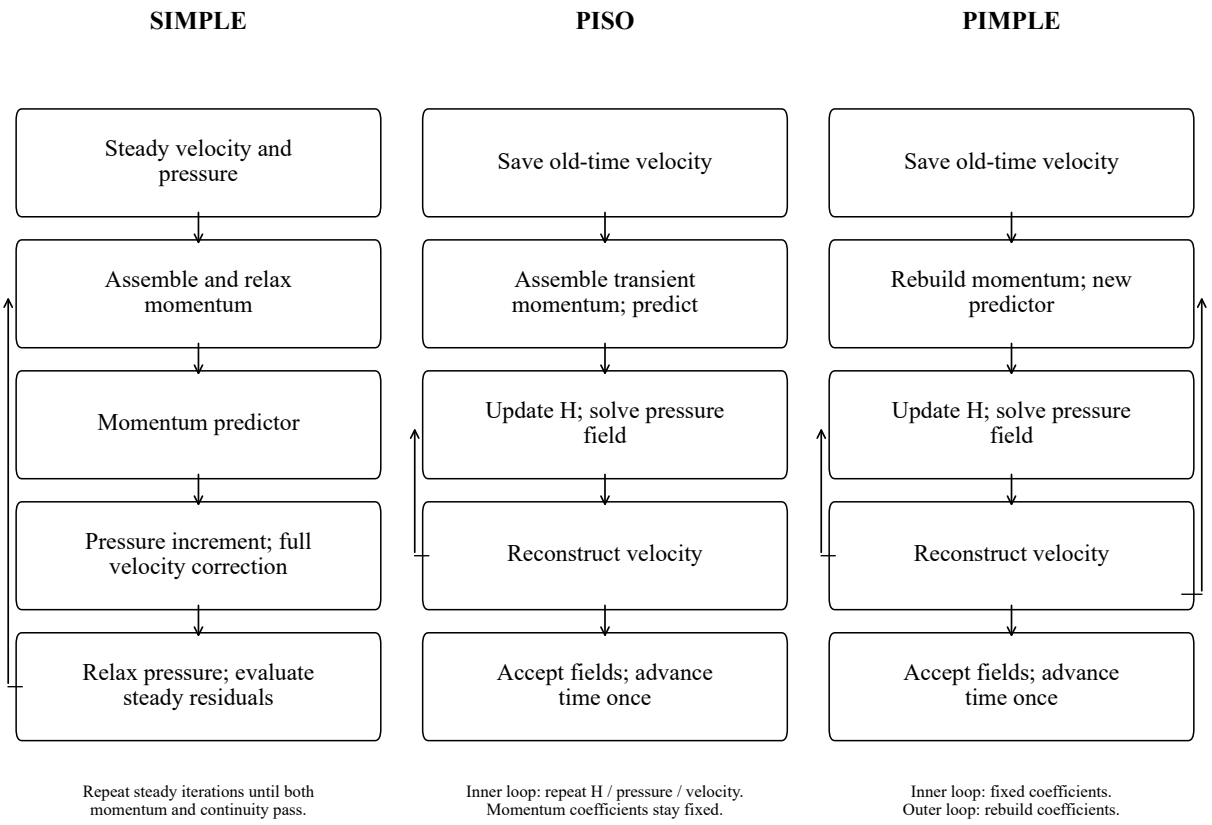
It is different from the residual of the frozen predictor matrix. All final mass defects are below $10^{-15}$.

For the exact constrained solution set $q_1 = q_2 = s$ and add the two momentum rows:

$$(9 + 2s)s = 2, \qquad s = \frac{\sqrt{97} - 9}{4} = 0.2122144504, \qquad p = s/2 - 1.$$

The table shows the benefit of coefficient updates for this example. Actual cavity efficiency must be judged using its own measured equation defects, accuracy and solve costs.

# 14 Algorithm loops and computational work




The nested loops act on one target problem at the current physical step. The diagram distinguishes steady SIMPLE iterations from PISO and PIMPLE time integration.

Let $N_i$ be the number of pressure correctors and $N_o$ the number of outer correctors. Each predictor solves horizontal and vertical momentum separately.

| Work unit | SIMPLE iteration | PISO step | PIMPLE step |
|---|---|---|---|
| Momentum assemblies | 1 | 1 | $N_o$ |
| Component momentum solves | 2 | 2 | $2N_o$ |
| Pressure solves | 1 | $N_i$ | $N_oN_i$ |
| Physical time advances | 0 | 1 | 1 |

For three outer and two inner correctors, PIMPLE uses three assemblies, six component momentum solves and six pressure solves per physical step. A direct linear solver may factor a pressure matrix once and reuse it within an inner loop, while solving different right-hand sides.

An iteration count alone cannot compare these costs. The retained runs use fixed corrector counts, rather than an adaptive inner tolerance. A steady stopping test is applied outside these operations.

# 15 Convergence, accuracy and the meaning of a residual

**Cell continuity.** The mass-balance diagnostic is $\max_P |r_{c,P}|$: the largest absolute continuity residual among all pressure cells. It is evaluated from face velocities, not from an interpolated picture of cell-center velocity.

**Momentum defect.** For a steady target, the implementation rebuilds the unrelaxed spatial coefficients using final velocity and evaluates the original momentum equations. For a transient step, a nonlinear step defect includes the same fixed old-time source:

$$r_{\text{step}} = A_{\text{space}}(q^{n+1})q^{n+1} - b_{\text{wall}} + \frac{q^{n+1} - q^n}{\Delta t} + Gp^{n+1}.$$

Small mass defect alone does not make this vector small, as the PISO numerical example demonstrated.

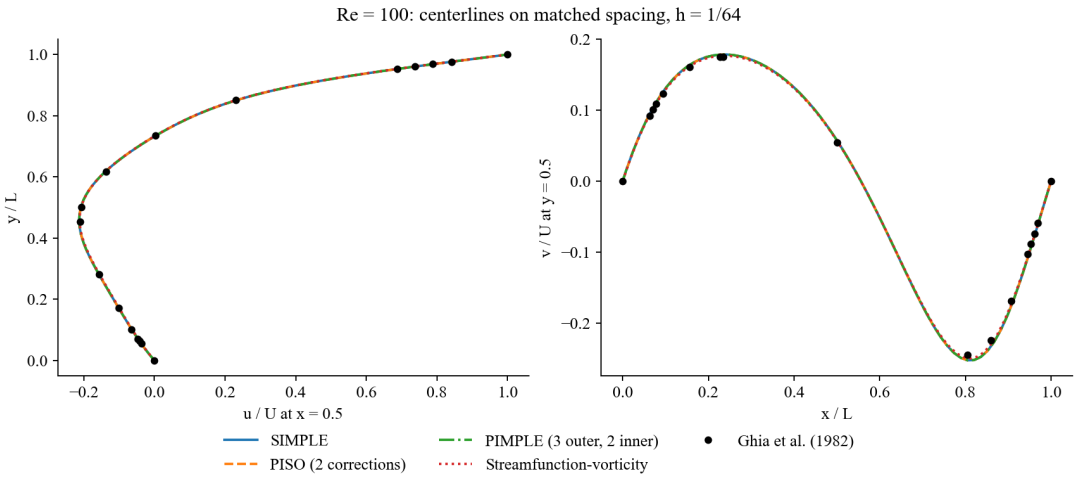
**Accuracy.** Agreement with independent reference velocities and stability under spatial refinement address solution accuracy. Ghia centerline tables validate selected velocity profiles; they do not validate pressure. A converged coarse-grid calculation can still be inaccurate.

**Transient accuracy.** Corrector loops reduce algebraic errors inside a step. Temporal error is assessed separately by reducing $\Delta t$ and comparing at matched physical times. Steady cavity agreement alone does not validate startup or oscillatory flow.

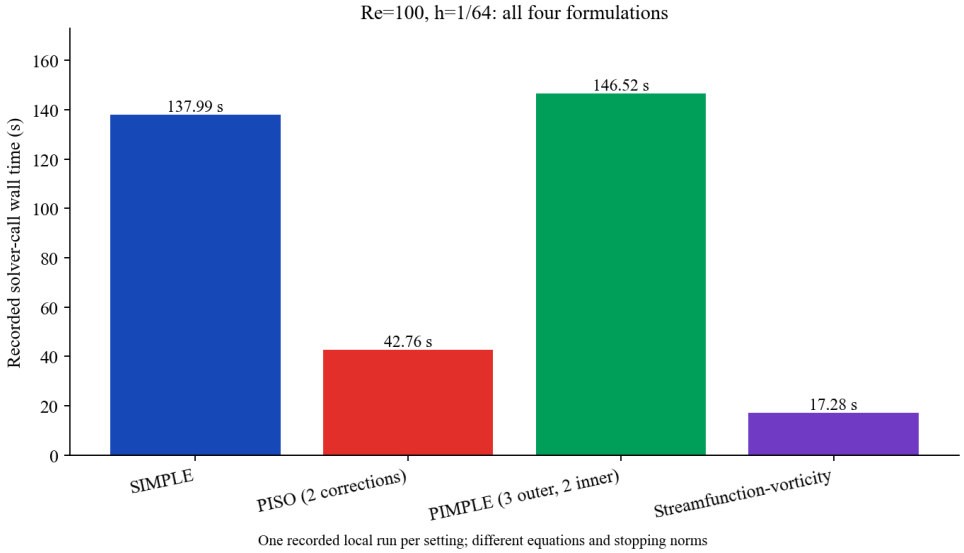
**Streamfunction–vorticity comparison.** Week 1 advances vorticity and solves streamfunction. Its native update norm is different from the finite-volume momentum defect. Their convergence histories use separate axes; common velocity errors and wall-clock times provide cross-formulation comparisons. $N$ finite-volume cells correspond to $N+1$ finite-difference nodes at matching spacing $h = 1/N$.

**Higher Reynolds numbers.** Stronger convection and thinner layers can require more cells and more work. More pressure corrections cannot recover spatial structures missing from the mesh. Refinement and higher-*Re* runs can take substantially longer and need more memory; the supplied coarse cases are the starting point for laptop or Colab use.

# 16 Recorded cavity accuracy and runtime at $Re = 100$



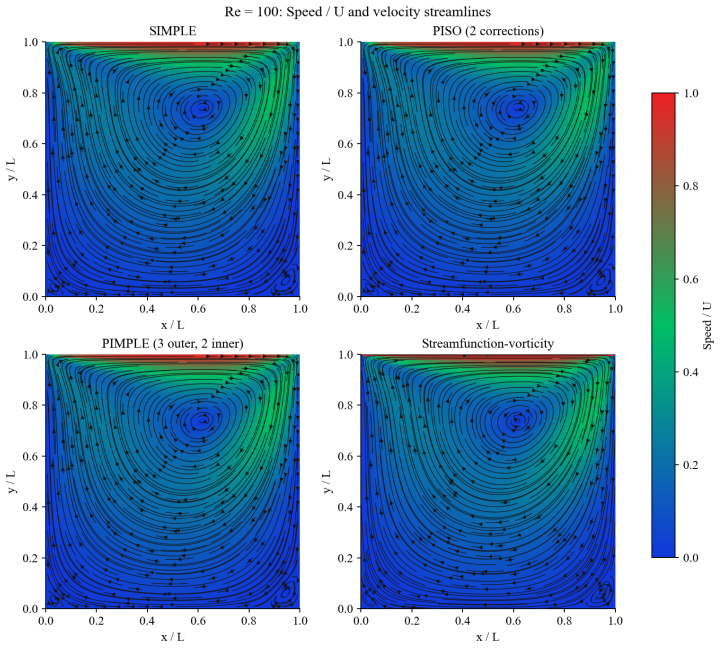

The Ghia velocity markers are independent reference samples. The three finite-volume coupling methods use the same spatial equations. Their converged fields agree closely; differences from the finite-difference formulation also include spatial-discretization differences. The displayed 32/64 finite-volume cell counts correspond to 33/65 finite-difference nodes.



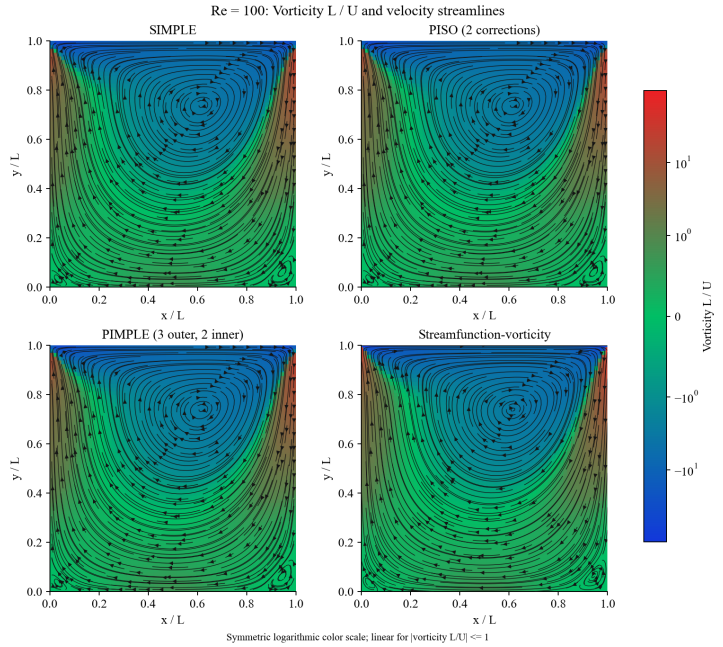

The bars include all four formulations and use recorded solver-call seconds. The finite-difference reference has its own stopping norm and recovers pressure afterward. These measurements depend on the host, implementation, mesh, relaxation and stopping targets; they do not establish a universal fastest method.

The finite-difference centerline errors decrease on refinement in this retained comparison. The finite-volume Ghia errors are not monotone between the two supplied grids. That behavior is retained rather than interpreted as proof of grid independence. More resolution and reference evidence are needed for an asymptotic accuracy claim.

# 17 Velocity structure and vorticity

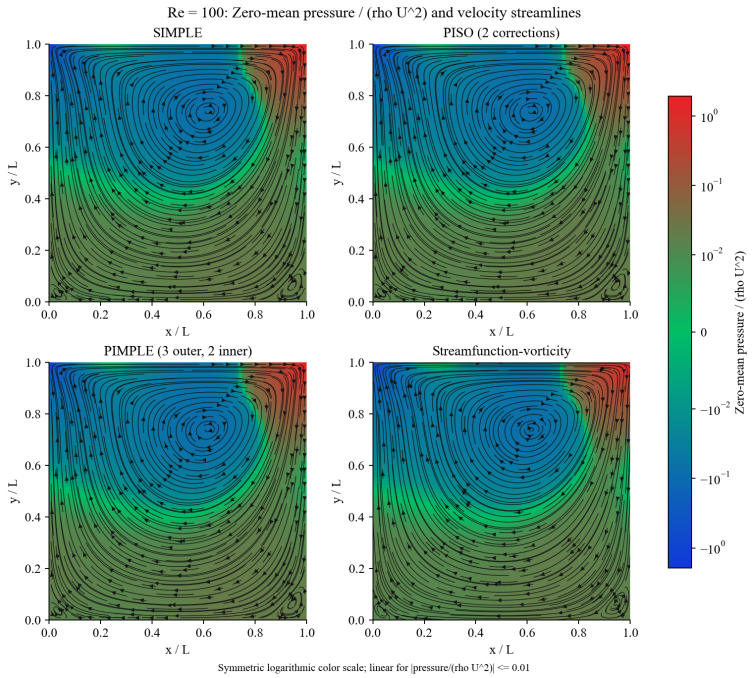



Speed is $|\boldsymbol{u}| = \sqrt{u^2 + v^2}$ and signed vorticity is $\omega = \partial v/\partial x - \partial u/\partial y$. The moving lid drives a clockwise primary circulation and a lower-cavity return flow. Black streamlines follow computed velocity; they are not vorticity contours. Small corner eddies depend on resolution and convergence as well as display density.

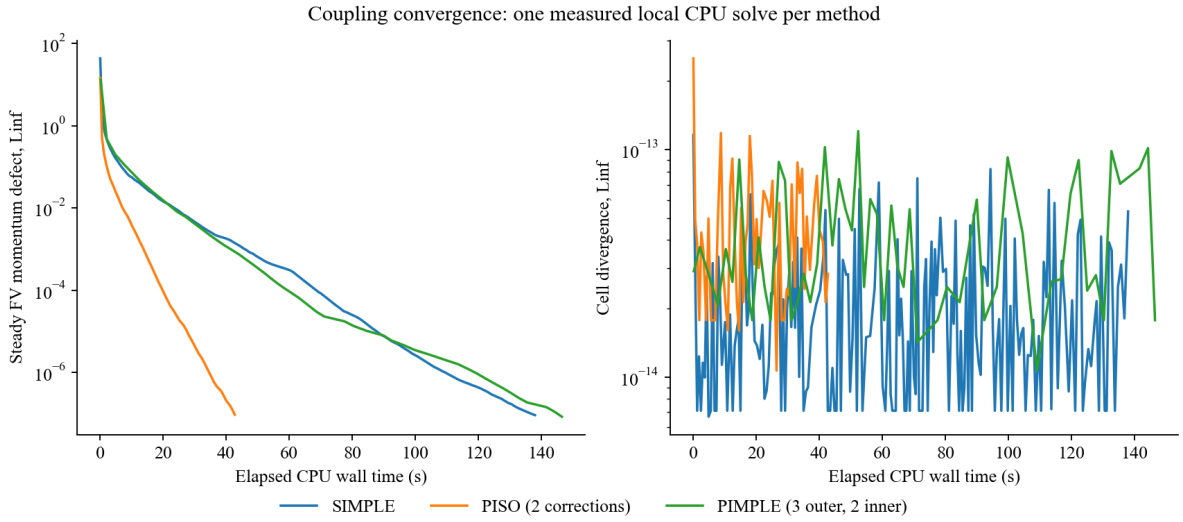



Signed fields use a common symmetric-logarithmic color scale across methods, resolving interior values while retaining strong wall gradients. Colorbar values remain physical nondimensional field values. Similar converged fields are expected when the same discrete steady equations are solved; the comparison is useful for checking consistency.

# 18 Pressure and native convergence histories

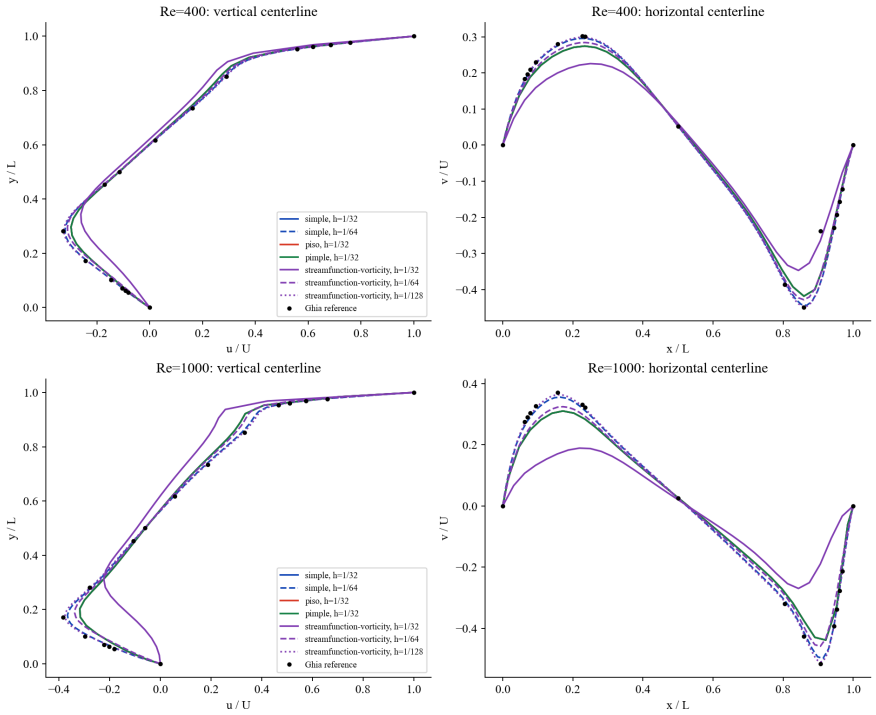



All pressures are reported with zero mean. Pressure gradients, rather than the arbitrary mean, affect velocity. Differences near the discontinuous lid corners should not be interpreted as reference-pressure validation. Ghia supplies velocity data, not pressure data.

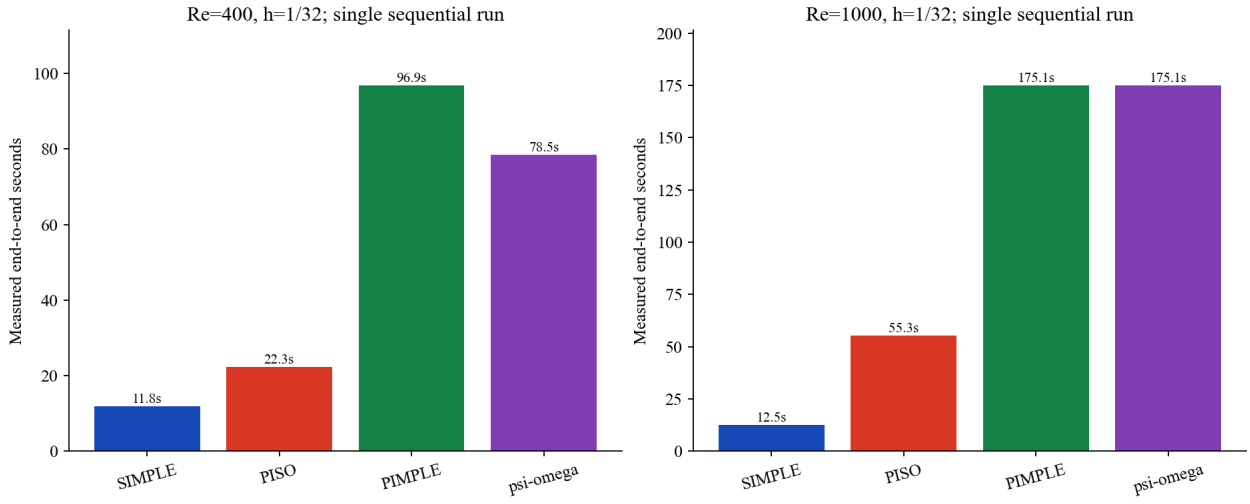



The finite-volume histories evaluate momentum and mass balance. The streamfunction solver's native vorticity-update norm is kept separate in the notebook because it is a different quantity. Timing comparisons still include that formulation. A small relaxed update alone is not a momentum-equation convergence test.

# 19 Accuracy and recorded cost at $Re = 400$ and 1000

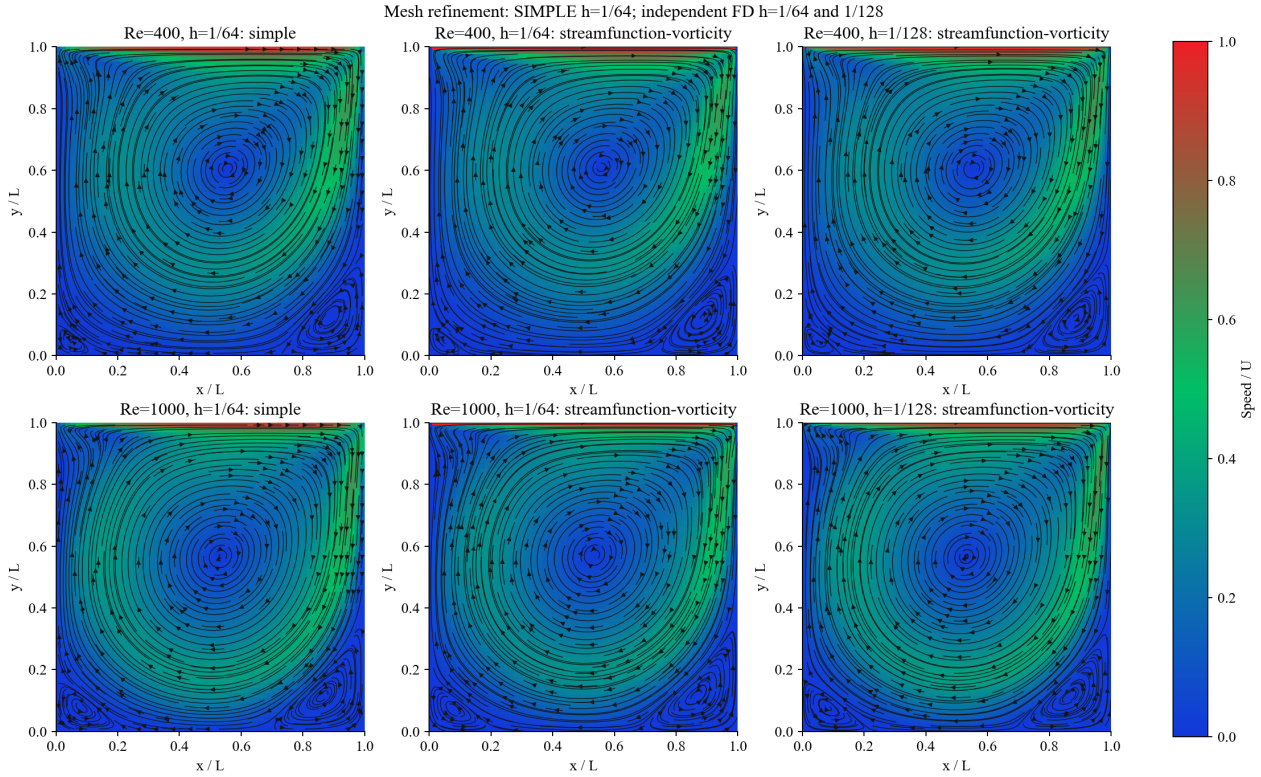



The retained study includes all four methods at $h = 1/32$, plus selected finer meshes. A converged coarse-grid streamfunction result can have large reference error. At $Re = 1000$, its 128-cell-spacing relative profile errors are approximately 2.0% for $u$ and 1.7% for $v$, using $\|f_{\text{computed}} - f_{\text{Ghia}}\|_2 / \|f_{\text{Ghia}}\|_2$ at the tabulated samples.




For an error vector $z$ over the reference samples, $\|z\|_2 = \sqrt{\sum_j z_j^2}$; $j$ labels a sampled location. These are recorded sequential local solver-call timings. Reference errors, mesh spacing and native stopping definitions accompany them. The $Re = 400$ all-point $v$ error retains the isolated published entry; a separately labeled sensitivity metric omits that sample without replacing its value.

# 20 Mesh refinement and recirculation

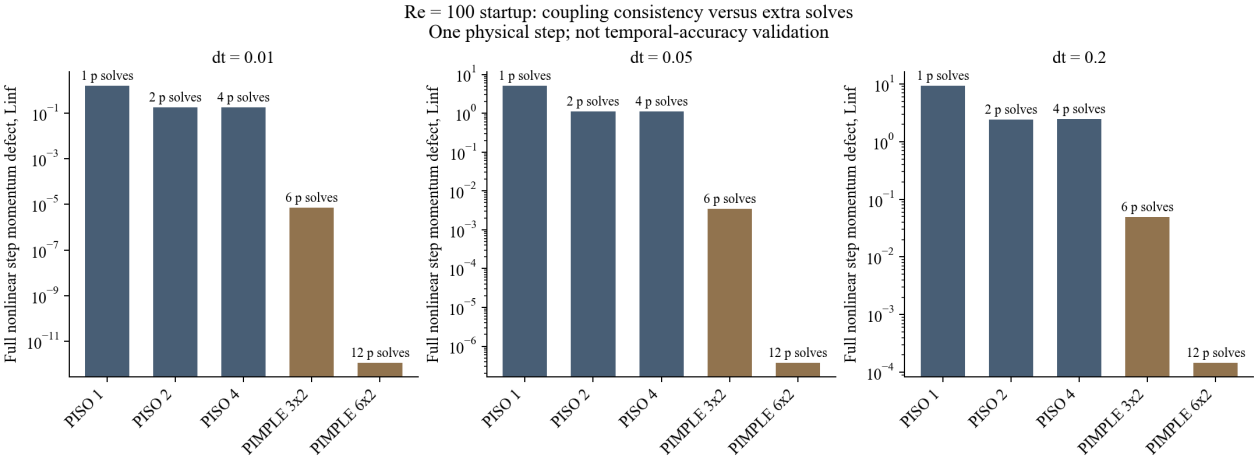



Each panel states Reynolds number, formulation and spacing. The sampled primary-vortex location and resolved corner circulation change with refinement. More inner pressure corrections cannot reproduce an unresolved boundary layer or secondary vortex.

The fine-grid study is selected rather than a complete four-method refinement matrix. A result at one mesh does not demonstrate asymptotic order. Native steady-equation checks, Ghia errors and recorded initial failed attempts remain available with the data.

**Computational cost.** Fine grids and higher Reynolds numbers may require substantially longer runtimes and more memory. Students should begin with the supplied coarse case and estimate cost before selecting 64 or 128 cells on a laptop or Colab runtime.

# 21 A measured nonlinear coupling effect in the cavity




Fifteen configurations start from rest at $Re = 100$ on 32 cells per side. Each takes one backward-Euler step. At each fixed $\Delta t$, the study varies inner pressure corrections and outer coefficient updates.

The nonlinear step diagnostic rebuilds convection from the final velocity, while the frozen-predictor diagnostic uses the coefficients assembled for the predictor. Extra PISO corrections can reduce pressure–velocity inconsistency without eliminating the error from frozen nonlinear coefficients. PIMPLE also updates those coefficients, with additional momentum assembly and solves.

All retained cases in this one-step study satisfy a cell-divergence defect below $10^{-9}$. The remaining nonlinear momentum defects distinguish the correction strategies. Their differences demonstrate algebraic coupling behavior for this setup, not the accuracy of a transient trajectory.

Each panel has its own logarithmic vertical range. For a temporal-accuracy study, reduce $\Delta t$ and compare at matched physical times. More correctors and a stable larger time step do not establish an accurate startup history or a universal wall-time advantage.

## 22 Notebook controls and student analysis

The notebook presents the same derivations and runnable algebraic examples, followed by retained CFD results. Fresh CFD runs are opt-in so the default lesson can be inspected without repeating every expensive calculation.

| Control | Purpose |
|---|---|
| `RUN_NEW` | Fresh baseline calculation at $Re = 100$ |
| `RUN_HIGHER_RE` | Fresh experimental higher-$Re$ calculation |
| `HIGHER_METHOD` | One method or all four formulations |
| `HIGHER_CELLS` | Number of cells per side / matched FD spacing |
| `RUN_STREAMFUNCTION` | Independent FD refinement experiment |

**Conservation.** Derive the four-face cell pressure equation and verify the two-cell example. Explain why a zero global flux does not guarantee zero cell imbalance. Add a constant pressure offset and check that velocities do not change.

**PISO.** Reproduce both corrections of the frozen-matrix example. Identify which quantities change between correctors. Explain why zero divergence after the first correction does not imply zero momentum defect.

**PIMPLE.** Reproduce the three outer passes of the nonlinear example. Evaluate the nonlinear defect using coefficients rebuilt from the final velocity. Explain why the old-time source must stay fixed. Compare one outer pass with PISO.

**Cavity comparison.** Start at $Re = 100$ and matching mesh spacing. Record final cell divergence, unrelaxed momentum defect, both Ghia profile errors, runtime, and solve counts. Change one coupling setting at a time; keep spatial equations and final accuracy targets fixed.

**Higher Reynolds number.** At $Re = 400$ and 1000, separate spatial refinement from coupling changes. Retain failed configurations. State the cost before proposing a fine-grid run. Native FD and FV residual magnitudes must not be interpreted as the same error.

**Transient extension.** Specify a final physical time, run a sequence of halved time steps, and compare common field or phase errors. Keep the time scheme fixed when comparing coupling algorithms. The current steady SIMPLE iteration is not a transient trajectory.

# 23 References

The textbook treatments develop finite-volume conservation, staggered pressure–velocity coupling and pressure correction. The paper and official algorithm documentation provide the PISO origin and practical inner/outer-loop distinctions.

[1] S. V. Patankar (1980). *Numerical Heat Transfer and Fluid Flow*. Hemisphere/Taylor & Francis. Publisher. Foundational pressure correction, staggered grids and SIMPLE.

[2] H. K. Versteeg and W. Malalasekera (2007). *An Introduction to Computational Fluid Dynamics: The Finite Volume Method*, 2nd ed. Pearson. Publisher. Finite-volume transport and pressure–velocity coupling.

[3] J. H. Ferziger, M. Perić and R. L. Street (2020). *Computational Methods for Fluid Dynamics*, 4th ed. Springer. doi:10.1007/978-3-319-99693-6. Incompressible-flow solution methods and numerical errors.

[4] F. Moukalled, L. Mangani and M. Darwish (2016). *The Finite Volume Method in Computational Fluid Dynamics*. Chapter 15, "Fluid Flow Computation: Incompressible Flows," pp. 561–654. Chapter DOI. Staggered/collocated arrangements and SIMPLE/PISO derivations.

[5] R. I. Issa (1986). "Solution of the implicitly discretised fluid flow equations by operator-splitting." *Journal of Computational Physics* 62(1), 40–65. doi:10.1016/0021-9991(86)90099-9. Original PISO formulation.

[6] CFD Direct, *OpenFOAM User Guide v13*, sections 4.6.7–4.6.8. Solution and algorithm control. Pressure reference, correctors and outer correctors. The teaching solver's fixed-loop MAC implementation is specified in the preceding equations; software variants need not have identical defaults.

[7] OpenCFD, *Pressure–velocity coupling*. Official equation decomposition. Momentum diagonal and neighbor-source representation.

[8] U. Ghia, K. N. Ghia and C. T. Shin (1982). "High-Re solutions for incompressible flow using the Navier–Stokes equations and a multigrid method." *JCP* 48, 387–411. doi:10.1016/0021-9991(82)90058-4. Velocity benchmark samples.

The examples use a staggered grid. Collocated-grid interpolation and Rhie–Chow stabilization are separate topics and are not implemented by this solver.